

The topology of software risk in scientific models

6. Dynamic profiling validation

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1 Preliminary

```
# PRELIMINARY FUNCTIONS #####
#####

sensobol::load_packages(c("data.table", "tidyverse", "tidygraph", "igraph",
                          "scales", "parallel", "here", "sensobol",
                          "benchmarkme", "softwareRisk", "boot", "MatchIt",
                          "effsize"))

# Set seed -----

seed <- 123
set.seed(seed)

# Source all .R files in the "functions" folder -----

r_functions <- list.files(path = here("functions"), pattern = "\\..R$", full.names = TRUE)
lapply(r_functions, source)
```

2 Model registry

```
# MODEL REGISTRY #####

cg_dir <- "dynamic_validation/callgraph_csv/Callgraphs"
xm_dir <- "dynamic_validation/extra_models_csv"

model_registry <- data.table(
  model = c("ORCHIDEE", "PCR-GLOBWB", "HYPE", "MHM", "SWAT"),
  format = c("two_file", "two_file", "combined", "combined", "combined"),
  static = c(file.path(cg_dir, "13_ORCHIDEE-MAN-2025_ORIGINAL", "callgraph_edges.csv"),
              file.path(cg_dir, "14_PCR-GLOBWB_model", "callgraph_edges.csv"),
              file.path(xm_dir, "HYPE"),
              file.path(xm_dir, "mHM"),
              file.path(xm_dir, "SWATplus")),
  runtime = c(file.path(cg_dir, "13_ORCHIDEE-MAN-2025-AND IOIPSL", "runtime_callgraph_edges.csv"),
               file.path(cg_dir, "14_PCR-GLOBWB_model", "runtime_callgraph_edges.csv"),
               file.path(xm_dir, "HYPE"),
               file.path(xm_dir, "mHM"),
               file.path(xm_dir, "SWATplus")),
  cc = c("ORCHIDEE", "PCR-GLOBWB", "HYPE", "MHM", "SWAT")
)
```

3 Static risk scores and runtime join

```
# BUILD P_k AND JOIN RUNTIME E_k #####

joined <- lapply(seq_len(nrow(model_registry)), function(i) {
  ed <- load_edges_fun(model_registry$static[i], model_registry$runtime[i],
    model_registry$format[i])
  jn <- join_runtime_fun(build_static_fun(ed$el, model_registry$cc[i]), ed$rt)
  jn[, model := model_registry$model[i]]
})
names(joined) <- model_registry$model
```

4 Rank correlation between P_k and runtime intensity

```
# SPEARMAN P_k VS E_k WITH BOOTSTRAP CI #####

corr_dt <- rbindlist(lapply(joined, function(dt) {
  ex <- dt[any_exercised == TRUE & !is.na(P_k_q50)]
  bo <- boot(ex, function(d, i)
    suppressWarnings(cor(d$P_k_q50[i], d$Ek_sum[i], method = "spearman")), R = 1000)
  ci <- boot.ci(bo, type = "perc")$percent[1, c(4, 5)]
  data.table(model = dt$model[1], n_exercised = nrow(ex),
    spearman = bo$t0, ci_lo = ci[1], ci_hi = ci[2])
}))

corr_dt
```

##	model	n_exercised	spearman	ci_lo	ci_hi
##	<char>	<int>	<num>	<num>	<num>
## 1:	ORCHIDEE	913	0.4971958	0.4406202	0.5473802
## 2:	PCR-GLOBWB	1412	0.2390355	0.1842273	0.2902270
## 3:	HYPE	7081	0.4175907	0.3949640	0.4391849
## 4:	MHM	698	0.3950496	0.3218790	0.4606601
## 5:	SWAT	3905	0.2893356	0.2605129	0.3216276

5 Matched-frequency contrast

```
# HIGH VS LOW P_k PATHS MATCHED ON E_k, HOPS AND SIZE #####

matched_dt <- rbindlist(lapply(joined, function(dt) {
  dt <- dt[!is.na(P_k_q50)]
  qhi <- quantile(dt$P_k_q50, 0.90); qlo <- quantile(dt$P_k_q50, 0.10)
  cvm <- median(dt$CV_k, na.rm = TRUE)
  dm <- dt[any_exercised == TRUE]
  dm[, grp := fifelse(P_k_q50 >= qhi & CV_k < cvm, 1L,
    fifelse(P_k_q50 <= qlo, 0L, NA_integer_))]
})
```

```

dm <- dm[!is.na(grp)]
dm[, stmt := pm_cc * hops]
mo <- matchit(grp ~ Ek_sum + hops + stmt, data = dm, method = "nearest",
             distance = "mahalanobis", replace = TRUE)
md <- match.data(mo)
rbindlist(lapply(c("pm_cc", "pm_indeg", "pm_btw"), function(mt) {
  a <- md[[mt]][md$grp == 1]; b <- md[[mt]][md$grp == 0]
  data.table(model = dt$model[1], metric = mt, n_pairs = sum(md$grp == 1),
             A = effsize::VD.A(a, b)$estimate,
             wilcox_p = suppressWarnings(wilcox.test(a, b)$p.value))
}))
}))

dcast(matched_dt, model + n_pairs ~ metric, value.var = "A")

```

```

## Key: <model, n_pairs>
##      model n_pairs pm_btw      pm_cc pm_indeg
##      <char>  <int>  <num>      <num>    <num>
## 1:      HYPE    967      1 0.8283351 1.0000000
## 2:       MHM    164      1 0.5487805 0.9314024
## 3: ORCHIDEE    145      1 0.9816092 0.7908046
## 4: PCR-GLOBWB   224      1 0.8928571 0.9955357
## 5:       SWAT   461      1 0.7830803 1.0000000

```

6 Complexity on rarely exercised paths

HIGH VS LOW P_k WITHIN THE RUNTIME-HEAVY DECILE

```

reverse_dt <- rbindlist(lapply(joined, function(dt) {
  ex <- dt[any_exercised == TRUE & !is.na(P_k_q50)]
  hot <- ex[Ek_sum >= quantile(Ek_sum, 0.90)]
  hot[, st := ifelse(P_k_q50 >= median(P_k_q50), "hi", "lo")]
  a <- hot[st == "hi", pm_cc]; b <- hot[st == "lo", pm_cc]
  data.table(model = dt$model[1], n = nrow(hot),
             A_cc = effsize::VD.A(a, b)$estimate,
             wilcox_p = suppressWarnings(wilcox.test(a, b)$p.value))
}))

reverse_dt

```

```

##      model      n      A_cc      wilcox_p
##      <char> <int>      <num>      <num>
## 1: ORCHIDEE   106 0.8454966 8.824840e-10
## 2: PCR-GLOBWB 249 0.7096452 1.080716e-08
## 3:      HYPE   722 0.1569432 2.638195e-57
## 4:      MHM    76 0.9168399 4.019649e-10
## 5:      SWAT   620 0.3788762 1.794384e-07

```

7 Session information

```
# SESSION INFORMATION #####
```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] C
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] parallel stats graphics grDevices utils datasets methods
## [8] base
##
## other attached packages:
## [1] effsize_0.8.1 MatchIt_4.7.2 boot_1.3-32 softwareRisk_0.2.0
## [5] benchmarkme_1.0.8 sensobol_1.2.0 here_1.0.2 scales_1.4.0
## [9] igraph_2.2.1 tidygraph_1.3.1 lubridate_1.9.4 forcats_1.0.1
## [13] stringr_1.6.0 dplyr_1.1.4 purrr_1.2.1 readr_2.2.0
## [17] tidyr_1.3.2 tibble_3.3.1 ggplot2_4.0.3.9000 tidyverse_2.0.0
## [21] data.table_1.18.4
##
## loaded via a namespace (and not attached):
## [1] benchmarkmeData_1.0.4 gtable_0.3.6 xfun_0.55
## [4] ggrepel_0.9.6 lattice_0.22-7 tzdb_0.5.0
## [7] vctr_0.7.3 tools_4.5.2 Rdpack_2.6.6
## [10] generics_0.1.4 pkgconfig_2.0.3 Matrix_1.7-4
## [13] RColorBrewer_1.1-3 S7_0.2.2 ineq_0.2-13
## [16] lifecycle_1.0.5 compiler_4.5.2 farver_2.1.2
## [19] ggforce_0.5.0 graphlayouts_1.2.2 codetools_0.2-20
## [22] htmltools_0.5.9 yaml_2.3.12 pillar_1.11.1
## [25] MASS_7.3-65 cachem_1.1.0 viridis_0.6.5
## [28] iterators_1.0.14 foreach_1.5.2 tidysselect_1.2.1
## [31] digest_0.6.39 stringi_1.8.7 polyclip_1.10-7
## [34] rprojroot_2.1.1 fastmap_1.2.0 grid_4.5.2
## [37] cli_3.6.6 magrittr_2.0.5 ggraph_2.2.2
## [40] withr_3.0.3 backports_1.5.0 timechange_0.3.0
## [43] rmarkdown_2.30 httr_1.4.8 otl_0.2.0
```

```
## [46] gridExtra_2.3      hms_1.1.4      memoise_2.0.1
## [49] evaluate_1.0.5      knitr_1.51     rbibutils_2.4.1
## [52] doParallel_1.0.17   viridisLite_0.4.3  rlang_1.2.0
## [55] Rcpp_1.1.1-1.1      glue_1.8.1     tweenr_2.0.3
## [58] R6_2.6.1
```

```
## Return the machine CPU -----
```

```
cat("Machine:    "); print(get_cpu())$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```
## Return number of true cores -----
```

```
cat("Num cores:  "); print(detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```
## Return number of threads -----
```

```
cat("Num threads: "); print(detectCores(logical = TRUE))
```

```
## Num threads:
```

```
## [1] 10
```